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Right to Be Forgotten: Implementation Challenges in the Digital Age

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Right to Be Forgotten: Implementation Challenges in the Digital Age

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Abstract—Right to Be Forgotten (RTBF) refers to the entitlement of an individual to have data related to them be deleted so that it is undiscoverable by third parties through online resources or search engines. This has become a bigger challenge over time with the continuously growing shift to the digital world that we currently live in because of how easy it is to find information about someone through a quick search on social media, Google, and other search engines, and the fact that these platforms store your data – sometimes without the permission of the individual this data is about – making it difficult to enforce RTBF across global platforms and creating major ethical concerns. The objective of this study is to further explain and highlight how the deletion of personal data is done online and how effective it is, along with the challenges of implementing this crucial right as we keep on finding ways to make everything digital, as well as discussing the different ethical issues that occur due to the secrecy of how our personal data is handled and who has access to it.

Keywords—right to be forgotten, general data protection regulation, cloud, artificial intelligence, digital privacy, ethical issues, privacy laws, compliance, deontological ethics, utilitarianism, virtue ethics

I. INTRODUCTION

A. Background and Context

In the early 2000s, there was a noticeable, rapid growth in internet usage, and most online platforms such as search engines and social media platforms began to store large amounts of personal data about users. If someone were to make a post online or when news articles were published, this information is stored and accessible even when it is no longer relevant or accurate. People are usually unaware of how their data is stored or for how long it is saved for and visible to third parties. This issue created major privacy concerns as individuals felt like they had almost no control over their personal information.

The Right to Be Forgotten (RTBF) became a need due to all these concerns. Its seriousness has increased significantly after the Google Spain v. AEPD case, one of the most known and famous cases related to privacy in Europe. The Google Spain case was raised after a Spanish man requested that Google deletes old news articles and headlines regarding the debt he was in years ago, claiming the information is no longer relevant and that it only harms his privacy and reputation now. Google refused to delete those articles as they are only a source to view these articles and are not responsible for their contents. This case went to the European Court where it was decided that people should have the right to ask search engines and social media platforms to delete personal information that is outdated,

inaccurate, or too private as it is more important to maintain privacy than feed public interest. After this case, the EU stated that there must be a law that clearly gives people the right and power to have their data deleted, resulting in the addition of RTBF to the EU's General Data Protection Regulation (GDPR), specifically article 17 [1].

Companies started using personal data for marketing purposes. Modern technologies were now allowing business to collect vast amounts of personal information from consumers, often without them being informed about how their data will be used [2]. Many companies use the data of their consumers in ways that are not explained beforehand or consented to, raising ethical concerns including lack of transparency and uninformed consent [2]. Situations like this are a root cause of the difficulty for people to protect their own privacy, highlighting the need for privacy laws, such as RTBF.

B. Problem Statement

Many people have their personal information stored online for long periods of time. This information may be old, wrong, or harmful, which can negatively affect a person's reputation, job opportunities, and most importantly their privacy. The problem worsens when online platforms refuse to have this information deleted, or when they allow deletion after a time-consuming or unclear process. With new technologies such as cloud storage and Artificial Intelligence (AI) emerging, data is saved in newer locations and is now harder to delete than ever because we can never be sure that personal information is still stored in hidden locations. Because of this, individuals lack the confidence of being able to control their own privacy as they are unable to protect it in the digital world [2].

C. Significance of the Study

This study is of high importance because it is becoming harder for people to manage their personal data with the evolution of new technologies. Many people are unaware of how long their information is accessible online or how it is being used by companies. RTBF is an important tool because it provides individuals with a way of having information that no longer represents them deleted. Ethically speaking, this study matters because it deals with the basic idea of personal autonomy. Personal data being accessible without clear consent violates the fundamental right to privacy [3]. People should have control over their own personal information and how it appears online. Understanding the challenges around RTBF helps in

showing where current systems fail to protect individuals and why stronger ethical guidelines are needed.

Overall, this research is significant because it encourages better privacy protection, clearer data-handling policies, and more responsible practices from organizations. It also helps build awareness among users who do not know they have the right to request deletion of their data, contributing to a safer and more trustworthy digital environment.

D. Research Objectives

- 1) Explain the importance of the Right to Be Forgotten (RTBF) in protecting privacy and personal data.
- 2) Examine the challenges individuals face when attempting to remove personal data from search engines and online platforms.
- 3) Analyze laws and regulations that grant the right to data erasure, including GDPR and its role in safeguarding privacy.
- 4) Identify the key stakeholders affected by RTBF, such as users, companies, digital platforms, and governments.
- 5) Evaluate the responsibilities of these stakeholders in protecting privacy, handling personal data ethically, and ensuring compliance with relevant legal frameworks.

II. LITERATURE REVIEW

Research on the Right to Be Forgotten (RTBF) has developed rapidly since the Court of Justice of the European Union (CJEU) recognized the right in *Google Spain v. AEPD* (2014), stating that individuals can request search engines to remove links that are “inadequate, irrelevant, or no longer relevant” [4]. This ruling became the foundation for RTBF in Europe and later influenced Article 17 of the General Data Protection Regulation (GDPR), which formally defines the right to erasure and outlines when data controllers must delete personal information [5]. As legal scholars note, a series of CJEU decisions — from *Google Spain* to later GDPR-related judgments — gradually shaped RTBF into a more structured and predictable legal right [13].

A major theme in the literature is that RTBF is much harder to implement in practice than it appears in law. Scholars and regulators agree that online data is highly distributed: the same piece of information can appear on multiple websites, archives, caches, and mirrored servers [6], making complete removal technically challenging. Search engines can delist links, but they cannot directly delete the original content hosted elsewhere, which limits the effectiveness of RTBF in a highly interconnected digital ecosystem [7], [8]. As a result, several researchers argue that RTBF today functions more like a “right to obscurity” rather than true erasure, because data traces can reappear through caches, republication, scraping, or automated aggregation tools [9].

Another major topic is the conflict between legal systems. Europe strongly protects privacy, while countries such as the United States prioritize freedom of expression, public records, and open access [10]. These differences make global enforcement difficult. The CJEU made this clear in *Google v. CNIL* (2019), ruling that search engines do not need to apply RTBF worldwide, but only within the EU domain [11]. Scholars argue that this decision confirms that

RTBF is context-dependent and cannot operate as a universal global rule.

Recent studies also highlight inconsistencies in decision-making. Data Protection Authorities (DPAs), courts, and search engines sometimes interpret “public interest,” “relevance,” and “sensitivity” differently, which can lead to different outcomes even in similar cases [12]. This variation shows that RTBF assessments are not always predictable and depend on institutional judgment.

A growing area of literature focuses on how RTBF applies to modern digital systems, especially artificial intelligence and large-scale data processing. Researchers explain that deleting a file from a database does not guarantee that the system has actually “forgotten” the information, because traces may exist in backups, logs, archives, or machine-learning models that were trained using the data [8]. This has led to discussions about technical mechanisms that support RTBF in practice. One key approach is **machine unlearning**, a technique designed to remove the influence of a specific person’s data from a model without retraining it from scratch. Early work by Cao and Yang demonstrated that systems can be designed to efficiently “forget” selected training records, making the model behave as if the data had never been included [22]. More recent surveys categorize the main unlearning methods, highlight open challenges, and emphasize that the field still lacks standardized tests to confirm that forgetting has truly happened [23].

Another technical difficulty identified in the literature concerns backups and archival systems. Many organizations store long-term backups for security or regulatory reasons, and these backups may contain data that should have been erased. Studies argue that unless backup procedures are updated, RTBF cannot be fully guaranteed, because deleted data may still exist in older snapshots or disaster-recovery copies [24].

Researchers also investigate the real-world effects of GDPR and RTBF on organizations and users. Empirical studies show that companies have increased their compliance efforts, but still face difficulties with cross-border requests, legacy systems, and interpreting vague legal terms [25]. From the user perspective, some individuals feel more empowered by the right, while others remain uncertain about what RTBF can or cannot achieve.

A newer strand of work examines the security implications of forgetting. Research shows that even if data is removed or “unlearned,” certain attacks — such as membership inference or model inversion — might still reveal whether a person’s data influenced a machine-learning model. Therefore, forgetting must be combined with strong verification and security practices to ensure that deleted data cannot be reconstructed or exploited [26].

A. Integration of Additional Recent Research

More recent studies extend RTBF research into areas that earlier literature did not fully consider. Earlier work mainly focused on legal interpretations, search engine delisting, and conflicts with freedom of expression [4], [5], [10], [11]. However, newer research demonstrates that the digital environment has changed significantly. Modern data ecosystems now include large-scale machine-learning systems, distributed storage infrastructures, automated data

pipelines, and long-term backups, all of which complicate the technical execution of RTBF requests.

For example, recent work highlights that data used to train artificial intelligence models can continue to influence model outputs even after the original data source is deleted [8]. This introduces a new category of RTBF challenges that did not exist when the original Google Spain judgment was issued in 2014. Studies on machine unlearning provide one of the first technical solutions aimed at removing the impact of a specific individual's data from an AI model, rather than only deleting the file itself [22]. These contributions show that effective RTBF compliance now requires addressing both stored data and learned representations inside algorithmic systems.

Other recent research focuses on organizational and infrastructural challenges. Studies show that many companies still struggle with deleting personal data from backups, mirrored servers, and legacy information systems, especially when these systems were not originally designed with erasure in mind [24], [25]. This is a major shift from earlier RTBF research, which mainly considered deletion from public-facing websites or search indexes.

Additionally, recent security-focused work highlights that even after deletion or unlearning, sophisticated attacks might still detect whether a model previously used a person's data [26]. This adds an entirely new security dimension to the discussion, showing that RTBF is not only a legal and technical challenge, but also a cybersecurity concern.

B. Comparison of Different Perspectives

The literature reveals clear differences across ethical, legal, and technical perspectives on RTBF.

1) Ethical Perspective

Ethical discussions focus on how RTBF affects fairness, dignity, and collective memory. Supporters argue that individuals should not be permanently harmed by old or irrelevant information that continues to appear online, and that RTBF helps protect personal dignity by allowing people a second chance [4], [5].

On the other hand, some ethicists caution that deleting information may hide facts that are important for public accountability, especially in cases involving public figures or socially significant events [10], [11]. Ethical concerns also arise around inequalities: individuals with more resources may be more capable of requesting deletion, while others may struggle to exercise this right effectively [12]. This suggests that the ethical value of the RTBF depends on balancing individual privacy with the needs of society.

2) Legal Perspective

Legal scholarship views the RTBF as an important development in strengthening individual privacy rights online. The Google Spain ruling established that people may request the removal of outdated or irrelevant links, marking a major shift in digital privacy protection [4]. This principle became law under Article 17 of the GDPR, which formally defines the right to erasure and the duties of data controllers [5]. Many authors argue that these rules are necessary to give individuals real control over their digital identities.

However, legal researchers also warn that the RTBF can conflict with freedom of expression and access to

information. Libraries, journalists, and civil society groups note that erasing or delisting information may limit the public's ability to access important records [10]. Analyses of CJEU cases show that courts often reach different outcomes depending on how they balance privacy with public interest, which leads to inconsistent enforcement across Europe [12], [13].

3) Technical Perspective

Technical research emphasizes that fully applying the RTBF is difficult due to the way digital systems store and spread information. Once data is copied across servers, caches, and mirrored platforms, it becomes very hard to remove completely [6]. Even when search engines delete links, the original content usually remains available elsewhere [7], [8]. Further challenges arise when deleted information reappears through cached copies or reposting [9], [24].

To address this, researchers have explored machine unlearning, which aims to design systems that can forget or remove specific data when required. Some studies see this as a way to support RTBF compliance in modern technologies [23], while others warn that current unlearning methods are not yet secure and may still leak information [26]. Another line of work argues that RTBF would be easier to implement if systems followed privacy-by-design principles and minimized stored data from the beginning [25].

C. Identification of Gaps in the Literature

Despite extensive research, several gaps remain in the understanding and implementation of the Right to Be Forgotten:

1) Lack of verified technical standards for "complete forgetting."

While machine unlearning and deletion protocols have been proposed, the literature acknowledges that no widely accepted standard exists for verifying when data — and its influence on models, has been fully removed [23], [26]. Current methods vary in accuracy, cost, and reliability, leaving uncertainty about compliance in AI-driven systems.

2) Insufficient solutions for backups, archives, and legacy systems

Scholars highlight that backups and long-term archives continue to store personal data even after erasure requests are processed [24]. Yet, research on how to modify or redesign backup systems for GDPR compliance remains limited.

3) Limited empirical evaluation of real-world RTBF practices

While some empirical studies exist, the literature lacks large-scale, long-term evaluations of how organizations implement RTBF over time and how often erasure is incomplete or inconsistent [25].

4) Unresolved global enforcement and cross-border issues

The CJEU's ruling in *Google v. CNIL* confirms that RTBF is territorially limited within the EU, but studies show limited understanding of how RTBF interacts with non-EU legal systems, especially those prioritizing free speech, such as the U.S. [10], [11].

5) *Complex challenges around AI and LLMs*

Research identifies that modern AI systems can encode personal data in non-obvious ways, but the literature provides no clear consensus on how RTBF should apply to continuously trained, large-scale models [8]. The difficulty of locating, isolating, and removing a single individual's contribution from a model remains an open research problem.

6) *Inconsistency in decision-making frameworks*

Studies show that different Data Protection Authorities and search engines may reach different conclusions for similar cases, but there is limited research on standardizing the assessment of relevance, public interest, and proportionality [12].

III. ETHICAL ANALYSIS FRAMEWORK

A. APPLICATION OF ETHICAL THEORIES

1) *Utilitarianism (Greatest Good for the Greatest Number)*

From a utilitarian view, the Right to Be Forgotten (RTBF) seeks to maximize overall well-being by allowing individuals to remove outdated or harmful information. This protects dignity, mental health, and future opportunities. However, removing information may limit public access to important facts, especially about public figures or safety concerns. RTBF is therefore justified when it reduces individual harm without significantly reducing the public's ability to access meaningful information [27].

2) *Deontological Ethics (Duty-Based Approach)*

Deontological ethics focuses on duties and obligations. Search engines have a duty to respect privacy and autonomy, which RTBF supports by giving individuals control over their data. However, they also have a duty to uphold freedom of expression and public access to information. Ethical implementation requires fair evaluation, transparency, and consistency while balancing both duties responsibly [27].

3) *Virtue Ethics (Character-Based Approach)*

Virtue ethics emphasizes the moral character and intentions of those making decisions. In the context of RTBF, organizations and IT professionals should act with honesty, integrity, and fairness when handling removal requests. Decisions should reflect virtues such as respect for privacy, responsibility, and prudence, ensuring that actions are morally sound rather than solely rule-based or outcome-focused. Implementing RTBF ethically under virtue ethics requires fostering a culture where decision-makers consistently prioritize

ethical principles and the well-being of both individuals and society [27].

4) *Rights-Based Approach*

This approach emphasizes fundamental rights such as privacy, autonomy, and reputation. RTBF supports these rights by allowing individuals to request removal of outdated or irrelevant information. At the same time, society holds the right to access accurate, relevant data. Ethical RTBF implementation must balance both sides by prioritizing individual rights when information is no longer useful while preserving public access when it serves legitimate interest [27].

5) *Justice and Fairness Approach*

Justice requires equal and unbiased treatment of all RTBF requests. Ethical implementation must avoid decisions based on status, influence, or financial power. It must also prevent unfair censorship of information that benefits the public. A just RTBF process is transparent, consistent, and balanced to ensure both individual protection and societal fairness [27].

IV. STAKEHOLDER ANALYSIS

1) *IT Professionals*

IT professionals must balance ethical, legal, and technical responsibilities when implementing RTBF. They are responsible for designing secure systems that process requests accurately, prevent misuse, and protect user data. They must avoid algorithmic bias and ensure that automated or manual evaluation processes are fair. Technical staff also face the challenge of removing data across backups and distributed systems without compromising system integrity [28].

2) *Organizations (Search Engines, Platforms, Data Controllers)*

Organizations face operational, legal, and financial pressures related to RTBF compliance. They must process large volumes of requests, maintain documentation, and justify their decisions. They face conflicting expectations: protecting privacy while avoiding excessive censorship. Failure to comply can result in regulatory penalties, while over-compliance may draw criticism for restricting information. Organizations must invest resources in legal review, secure deletion systems, and long-term data governance [28].

3) *End Users (General Public)*

End users benefit from improved control over their digital identity. RTBF allows individuals to reduce harm from outdated or misleading information. However, users may also experience reduced access to information when search engines remove relevant public-interest data. Another challenge is that understanding the scope and limitations of

RTBF is difficult, creating confusion about what can or cannot be deleted [29].

4) Society at Large

Society must consider the broader impacts of RTBF on historical record-keeping, transparency, and democratic accountability. Excessive deletion may weaken the public's ability to make informed decisions, while insufficient protection may allow outdated information to unjustly stigmatize individuals. RTBF requires balancing societal benefits of open information with the need to allow individuals to move beyond their past [29].

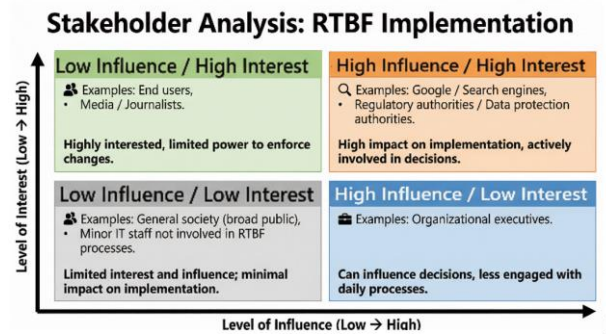


Fig. 1. Stakeholder Analysis: RTBF Implementation

V. RISK AND LIABILITY ASSESSMENT

A. Risk Assessment

1) Legal Risks

Organizations face significant legal risks if they fail to comply with RTBF regulations, particularly under GDPR. They must handle requests properly, document decisions, and demonstrate accountability. Conflicts between international laws—some prioritizing privacy, others prioritizing free speech—create additional legal uncertainty [30].

2) Financial Liabilities

RTBF compliance requires financial investment in staff training, legal consultation, data processing infrastructure, and security measures. Fines and lawsuits for noncompliance can be costly. Long-term liabilities include maintaining accurate deletion processes and updating systems to accommodate new regulations [30].

3) Reputational Risks

Improper RTBF handling can damage organizational reputation. Refusing to remove harmful information may be seen as privacy neglect, while removing too much may result in accusations of censorship or manipulation. Public trust depends on transparent, fair, and balanced decision-making [31].

4) Technical Vulnerabilities

Technical challenges include ensuring complete deletion across servers, backups, mirrors, and third-party systems. Organizations must prevent unauthorized removals or malicious RTBF requests. Scaling deletion processes to millions of users introduces risks of system errors, incomplete data removal, and algorithmic inconsistency [31].

TABLE I. RISK ASSESSMENT TABLE

ID	Risk Assessment			
	Risk Description	Impact	Likelihood	Risk Type
R1	Data Breach	High	High	Reputation + Technical
R2	Incorrect data deletion	High	High	Technical
R3	System downtime	Low	Medium	Financial + Technical
R4	Unauthorized access	High	Medium	Reputation + Technical
R5	Legal non-compliance	Medium	Low	Legal + Financial

1) R1 – Data breach

Someone might break into the system and steal private data [31].

Liability: Legal penalties, compensation claims.

2) R2 – Incorrect data deletion

The system might delete the wrong information by mistake [31].

Liability: Loss of critical records, user complaints.

3) R3 – System downtime

The system could stop working for a while and users can't access it [32].

Liability: Business disruption, revenue loss.

4) R4 – Unauthorized access

Someone who is not allowed might enter the system and see or change things [33].

Liability: Privacy violations, reputational damage.

5) R5 – Legal non-compliance

The system might not follow the law correctly, which could cause penalties [34].

Liability: Fines, regulatory sanctions.

VI. CASE STUDY ANALYSIS: MARIO COSTEJA GONZÁLEZ V. GOOGLE SPAIN (2014)

A. Case Study Overview

The following case study examines a landmark legal decision that illustrates the ethical and practical challenges of implementing the Right to Be Forgotten in the digital age. It demonstrates how conflicting interests between individual privacy and public access to

information are addressed through ethical decision-making [4] [21].

B. Identify the Facts

In 2014, Mario Costeja González requested that Google remove search results linking to a 1998 newspaper notice about his past financial debts. Although the debts were resolved, the information remained prominent in search results, negatively affecting his reputation. He argued that the information was outdated and no longer relevant. The Court of Justice of the European Union ruled in his favor, establishing the RTBF as a legal principle within the EU and requiring search engines to delist links that contain outdated or irrelevant personal information [4] [21].

C. Define the Ethical Issues

The ethical issues include whether individuals should control how outdated information about them appears online, whether removing search results restricts the public's right to information, and what duties search engines hold in balancing privacy rights with freedom of expression [21].

D. Identify Stakeholders

Stakeholders include Mario Costeja, Google as a data controller, the original newspaper that published the information, the public seeking information, and regulatory authorities responsible for enforcing data protection laws [4].

E. Consider Alternatives

1) Keep the search results unchanged

Leaving the links as they are would maintain full public access to information. However, the outdated content would continue to harm Costeja's reputation and personal life, even though the debts were already resolved.

2) Remove only the search link while preserving the original article

Delisting the Google search result protects Costeja's privacy while keeping the historical record intact. This balances individual rights with public access and minimizes harm.

3) Add clarifying context

Adding a note that the debts were resolved could reduce reputational damage. However, implementation is technically difficult, and users may ignore the clarification, limiting effectiveness.

4) Implement periodic review of information

Regularly reviewing links ensures that outdated or harmful content is identified and potentially removed over time. This requires significant resources and may not fully prevent temporary exposure to harmful information.

5) Limit removal to EU-based search domains

Restricting delisting to European versions of Google protects privacy for EU users, but individuals outside the EU can still access the information, reducing overall effectiveness.

F. Evaluate the Consequences

1) Keep the search results unchanged

Consequence: Costeja's outdated debts continue to appear prominently, damaging his reputation and professional opportunities.

Judgment: Weak Alternative, it fails to protect the individual and allows unnecessary personal harm.

2) Remove only the search link while preserving the original article

Consequence: The outdated information is hidden from search results, protecting Costeja's privacy, while the historical record remains accessible.

Judgment: Good Alternative, balances individual privacy with public interest effectively.

3) Add clarifying context

Consequence: Users might see that the debts were resolved, which could reduce harm, but many users may ignore the clarification, limiting its effectiveness.

Judgment: Partially good Alternative, helps mitigate harm but is inconsistent and less reliable.

4) Implement periodic review of information

Consequence: Outdated content is eventually identified and addressed, but harmful information may still be visible for long periods, and it requires significant resources.

Judgment: Moderate Alternative, better than doing nothing but not a complete solution.

5) Limit removal to EU-based search domains

Consequence: Users in the EU are protected, but people outside the EU can still access the information, leaving privacy only partially safeguarded.

Judgment: Partially bad Alternative, provides some protection but fails to fully achieve the purpose of RTBF.

G. Make a Decision

The most ethical decision is to remove the link from Google search results while preserving the original newspaper article.

H. Justify the Decision

Removing the search link balances privacy and public interest. It reduces personal harm while maintaining historical integrity and freedom of the press. The decision aligns with utilitarian ethics by maximizing overall well-being, with the rights-based approach by protecting individual privacy, and with justice by ensuring fair

treatment for individuals seeking removal of irrelevant or outdated information.

This case highlights that the Right to Be Forgotten extends beyond individual cases, setting a precedent for how search engines must balance privacy, public access, and ethical responsibilities. Effective RTBF implementation requires clear policies, robust technical systems, and fair evaluation processes, while also raising public awareness of its limitations [4] [21].

In conclusion, the Mario Costeja González case shows that RTBF is both a legal and ethical framework. Proper implementation protects individual privacy, maintains public trust, and ensures fair access to information, serving as a guide for responsible digital governance.

VII. RECOMMENDATIONS AND BEST PRACTICES

Implementing the Right to Be Forgotten (RTBF) in modern digital systems requires organizations to adopt technical, legal, and ethical strategies that ensure personal data can be erased reliably and responsibly. One of the most important best practices is the integration of machine unlearning mechanisms into data processing pipelines. Traditional deletion removes data from storage but does not erase its influence on trained models. Machine unlearning techniques, such as scrubbing, knowledge distillation, and selective retraining, allow systems to reverse the effect of specific data points without rebuilding entire models, which supports both efficiency and regulatory compliance [35]. These methods are especially valuable for large-scale systems where full retraining is computationally expensive and impractical.

Another essential recommendation is the adoption of verifiable deletion protocols. Many organizations claim they have erased user data, but users often have no technical means to confirm these claims. Recent frameworks propose statistical verification and influence-based auditing, enabling third parties to test whether a model still relies on deleted data [36]. Verifiable deletion not only supports transparency but also ensures that RTBF requests satisfy legal standards under GDPR Article 17. To strengthen this process, providers should maintain secure audit trails, documenting all deletion requests, the system components affected, and verification outcomes [37].

Furthermore, organizations should design systems that minimize data retention by default. Long-term storage of personal data increases privacy risks and complicates RTBF compliance. Privacy-by-design principles recommend using data minimization, short retention windows, and anonymization where possible. Studies show that many companies over-collect and store unnecessary data for extended periods, increasing exposure to security breaches and regulatory violations [38]. Implementing retention limits and automatic deletion policies ensures that personal data does not remain online longer than needed.

Low-latency deletion should also be a priority. Some systems take substantial time to process deletion requests, especially when data is distributed across cloud storage, replicated databases, and machine-learning models. Techniques such as incremental unlearning, partitioned model training (e.g., SISA), and targeted parameter updates help reduce the delay between a user request and its completion [39]. Faster processing not only improves user trust but also fulfills legal expectations of “without undue delay.”

Organizations must additionally consider fairness and equity when implementing unlearning. Research shows that removing data from certain groups more frequently can unintentionally distort model fairness and accuracy across demographic populations [40]. Best practices include auditing unlearning operations for fairness impacts, monitoring performance across sensitive groups, and selecting unlearning algorithms that minimize disparities.

Another key recommendation is improving organizational governance and internal policies. Companies should create clear RTBF procedures, outlining how requests are submitted, how long deletion takes, what systems are affected, and how residual data (e.g., backups or logs) is managed. Teams responsible for data compliance must receive continuous training on emerging privacy regulations, unlearning technologies, and ethical data-handling obligations [41]. Collaboration between legal, cybersecurity, and AI teams ensures that deletion practices are technically sound and legally aligned.

Users must also be empowered through better transparency and communication. Many individuals do not fully understand their data rights, how platforms handle their information, or what RTBF can and cannot accomplish. Organizations should provide accessible guidance explaining the deletion process, the technologies involved, and limitations such as model residual influence [42]. Clear communication strengthens public trust and aligns with global movements toward consumer-centric privacy frameworks.

Policymakers and regulators should work toward establishing clear standards for unlearning and deletion verification. Current laws define the right but provide limited technical guidance on implementation. Scholars argue that standardization of verification metrics, acceptable unlearning latency, and influence thresholds would ensure more consistent compliance across industries [43]. Regulatory bodies could also promote certification schemes that evaluate whether organizations meet these technical and ethical RTBF requirements.

By integrating these practices into system design, organizational policy, and regulatory frameworks, stakeholders can support a more effective and ethically grounded implementation of the Right to Be Forgotten, one

that addresses modern challenges in digital privacy, large-scale data processing, and machine learning.

VIII. CONCLUSION

The Right to Be Forgotten (RTBF) represents a critical advancement in protecting individual privacy in the digital era. As this study demonstrates, the implementation of RTBF is not merely a legal requirement but also an ethical and technical challenge that intersects with personal autonomy, public interest, and modern digital infrastructures. By examining the Mario Costeja González case and evaluating the responsibilities of stakeholders, it becomes clear that effective RTBF enforcement requires careful consideration of both individual rights and societal benefits.

Technological complexities, such as distributed storage, backups, and machine-learning models, pose significant obstacles to fully deleting personal data. Techniques like machine unlearning and verifiable deletion protocols are promising solutions, yet they require continuous refinement and standardization to ensure reliability and compliance across organizations. Furthermore, ethical considerations—ensuring fairness, transparency, and non-discrimination—must guide the development and deployment of deletion mechanisms, particularly in automated systems where inadvertent biases may arise.

From a governance perspective, organizations must implement clear policies, foster cross-departmental collaboration, and maintain detailed audit trails to uphold accountability. At the same time, users must be educated on their rights and the practical limitations of RTBF to foster trust and engagement. Regulatory bodies play a crucial role in defining standards, verification metrics, and certification frameworks, thereby ensuring consistency and legal alignment across jurisdictions.

In conclusion, the RTBF is more than a procedural obligation; it is a multidimensional framework that combines law, technology, ethics, and governance to protect personal data in the digital age. Its successful implementation strengthens digital trust, empowers individuals, and encourages responsible data stewardship by organizations. As technology continues to evolve, continuous research, updated legal frameworks, and practical technical solutions will be essential to fully realize the potential of the Right to Be Forgotten while balancing privacy with the broader public interest.

IX. GROUP CONTRIBUTION STATEMENT

All members of the group contributed extensively and collaboratively to the research, analysis, and writing of this paper. Nada took the lead in drafting the introduction and abstract, setting the tone and framework for the study. Lana conducted a thorough literature review, ensuring that the paper was grounded in current and credible scholarly sources. Lara developed the figures and case study analysis, providing visual clarity and practical examples to support

the research findings. Beyond these individual contributions, all members actively participated in researching, writing, editing, and refining the content to ensure accuracy, coherence, and a high standard of quality. The completion of this paper reflects the combined dedication, effort, and teamwork of the entire group.

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